25901002 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOUR - I

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer **All** Questions

Each Question carries **1** mark

1. The person who treat people with emotional distress or troublesome behaviors, sometimes using drugs or other medical procedures is _____. [U] / [CO1]
2. Operant conditioning was first described by psychologist--------- [U] / [CO1]
3. The voltage measured when the inside of the cell is negative relative to the outside is known as the _____ potential. [K] / [CO2]
4. _____is an enzyme which break catecholamines [U] / [CO2]
5. The threshold for triggering an action potential is typically around _____ mV. [K] / [CO2]
6. Sensory neurons (or afferent neurons) carry impulses from sensory receptors toward the _____. [K] / [CO2]
7. _____ is the primary neurotransmitter involved in forebrain activity linked to reward mechanisms. [U] / [CO3, CO4]

8. The hypothalamus communicates with the autonomic nervous system to regulate involuntary functions such as _____ and _____. [U] / [CO3, CO4]

9. The left hemisphere of the brain is primarily responsible for _____ processing [K] / [CO3, CO4]

10. The _____ houses the cardiac, respiratory, and vasomotor control centers. [K] / [CO3, CO4]

[10x1 = 10]

Part B

Short Questions

Answer any **10** Questions

Each Question carries **2** marks

11. Write about the bodily factors that influence the mind. [U] / [CO1]

12. Define BMR. [K] / [CO1]

13. What are ependymal cells and explain its role in nervous system? [U] / [CO2]

14. What is the significance of K+ leak channels in resting potential? [U] / [CO2]

15. Explain the role of Ranvier's nodes in impulse transmission. [U] / [CO2]

16. Differentiate between short-term and long-term potentiation. [U] / [CO2]

17. Comment on the types of neuroglial cells. [U] / [CO2]

18. How does the parasympathetic nervous system help maintain homeostasis? [U] / [CO3, CO4]

19. Describe the physiologic changes happen during the stimulation of symapathetic fibres. [U] / [CO3, CO4]

20. Describe the role of the cranial nerves in the peripheral nervous system. [U] / [CO3, CO4]

21. What are the functions of the brainstem in the central nervous system? [An] / [CO3, CO4]

22. Describe the structure and function of the meninges in brain. [U] / [CO3, CO4]

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **5** Questions

Each Question carries **4** marks

23. Write a short note on different stages of human evolution?	[K] / [CO1]
24. Describe the legal system that regulates animal research in the U.S. and the role of the Institutional Animal Care and Use Committee.	[U] / [CO1]
25. Describe the stages of action potential with the help of a labeled diagram.	[U] / [CO2]
26. Discuss excitatory post synaptic potential.	[K] / [CO2]
27. How does the midbrain contribute to the regulation of movement and coordination?	[An] / [CO3, CO4]
28. Explain the role of the spinal cord in reflex action.	[U] / [CO3, CO4]
29. Give an account on the structure of neocortex.	[U] / [CO3, CO4]

[5x4 = 20]